Enrollment No.: _____

Darshan UNIVERSITY

योगः कर्मसु कौशलम्

Darshan Institute of Computer Application

BCA | Sem-5 | Summer-2025

Course Code	:	2104CS521	**Date**	:	14-05-2025
Course Name	:	Introduction to Advanced Technologies	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Define Cloud Computing. List various cloud service providers available in the industry.	**4**
	(B)	Explain characteristics of cloud computing.	**3**
		OR	
		Explain Feature of Cloud Computing.	
	(C)	Explain the cloud service model (IAAS, PAAS, SAAS).	**7**
		OR	
		List out types of Virtualizations. Explain anyone in detail.	
Q.2	**(A)**	Compare Augmented Reality and Virtual Reality with minimum two applications.	**4**
	(B)	Explain Visualization Techniques for Virtual Reality (VR).	**3**
		OR	
		Explain Mixed Reality with an example.	
	(C)	Explain challenges with AR and VR.	**7**
		OR	
		Explain application of AR/VR.	
Q.3	**(A)**	Define Testing, Training, Label, Prediction.	**4**
	(B)	Explain Supervised Machine Learning.	**3**
		OR	
		Explain Un-supervised Machine Learning.	
	(C)	Explain Machine Learning application.	**7**
		OR	
		Draw and explain machine learning life cycle with diagram.	

Q.4	**(A)**	Explain application of data science.	**4**
	(B)	Explain how to read Excel in pandas with example.	**3**
		OR	
		Explain how to read CSV in pandas with example.	
	(C)	Explain python ecosystem for data science.	**7**
		OR	
		Explain groupby() function in pandas with example.	
Q.5	**(A)**	Define concept of decentralization.	**4**
	(B)	Explain altcoins in brief.	**3**
		OR	
		Describe fundamental working of Proof of Stake consensus algorithm.	
	(C)	Explain types of blockchain.	**7**
		OR	
		What is cryptocurrency? Explain types of forking.	

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